Question 1

(a) Distance Calculations

Given data objects: $x : (5, 3, 7, 9)$ and $y : (7, 3, 6, 7)$.

1. **The Hamming distance between x and y**
   Hamming distance measures the number of positions where symbols differ (mismatches).

$$\begin{aligned} &(5 \neq 7) \to 1 \\ &(3 = 3) \to 0 \\ &(7 \neq 6) \to 1 \\ &(9 \neq 7) \to 1 \end{aligned} \quad \Rightarrow \quad \text{Total} = 1 + 0 + 1 + 1 = 3$$

   Answer: 3

2. **The Manhattan (L_1) distance between x and y** ($d_1(x, y) = \sum |x_i - y_i|$)

$$d_1(x, y) = |5 - 7| + |3 - 3| + |7 - 6| + |9 - 7| = 2 + 0 + 1 + 2 = 5$$

   Answer: 5

3. **The Euclidean (L_2) distance between x and y** ($d_2(x, y) = \sqrt{\sum (x_i - y_i)^2}$)

$$d_2(x, y) = \sqrt{(-2)^2 + 0^2 + 1^2 + 2^2} = \sqrt{4 + 0 + 1 + 4} = \sqrt{9} = 3$$

   Answer: 3

(b) The Problem with Unscaled Data in Clustering

Height: $\approx 1500 - 1900$ mm.
Eyelash length: $\approx 5 - 15$ mm.
Since height values are much larger, they will **dominate** the distance calculation. A small % change in height makes a larger contribution than a large % change in eyelash length. The algorithm will perceive eyelash length as irrelevant.
Solution: Normalise or **standardise** (Feature Scaling) before clustering using:
Min-Max Normalisation: Rescaling to $[0, 1]$.
Z-score Standardisation: Mean of 0, SD of 1.

(c) K-means Clustering Simulation

Initial Centres: C_1: $A(2, 10)$, C_2: $D(5, 8)$, C_3: $G(1, 2)$.

i. Assignment of Remaining Objects (Squared Euclidean Distance)

1. **Object B (2, 5):**
   To A: $(2 - 2)^2 + (5 - 10)^2 = 25$. To D: $(2 - 5)^2 + (5 - 8)^2 = 18$. To G: $(2 - 1)^2 + (5 - 2)^2 = 10$ (**Min**).
   $\to$ **Assign B to Cluster G.**
2. **Object C (8, 4):**
   To A: $36 + 36 = 72$. To D: $9 + 16 = 25$ (**Min**). To G: $49 + 4 = 53$.
   $\to$ **Assign C to Cluster D.**
3. **Object E (7, 5):**
   To A: $25 + 25 = 50$. To D: $4 + 9 = 13$ (**Min**). To G: $36 + 9 = 45$.
   $\to$ **Assign E to Cluster D.**
4. **Object F (6, 4):**
   To A: $16 + 36 = 52$. To D: $1 + 16 = 17$ (**Min**). To G: $25 + 4 = 29$.
   $\to$ **Assign F to Cluster D.**
5. **Object H (4, 9):**
   To A: $4 + 1 = 5$. To D: $1 + 1 = 2$ (**Min**). To G: $9 + 49 = 58$.
   $\to$ **Assign H to Cluster D.**
   Summary Step 1: Cluster A: $\{A\}$; Cluster D: $\{D, C, E, F, H\}$; Cluster G: $\{G, B\}$.

ii. New Cluster Centres

1. **Cluster A:** Centre $= (2, 10)$.
2. **Cluster D:** (Contains D, C, E, F, H)

$$x_{new} = \frac{5 + 8 + 7 + 6 + 4}{5} = 6, \quad y_{new} = \frac{8 + 4 + 5 + 4 + 9}{5} = 6 \quad \to \textbf{Centre: } (6, 6)$$

3. **Cluster G:** (Contains G, B)

$$x_{new} = \frac{1+2}{2} = 1.5, \quad y_{new} = \frac{2+5}{2} = 3.5 \quad \rightarrow \textbf{Centre: } (1.5, 3.5)$$

Question 2

(a) Hierarchical Clustering Evaluation

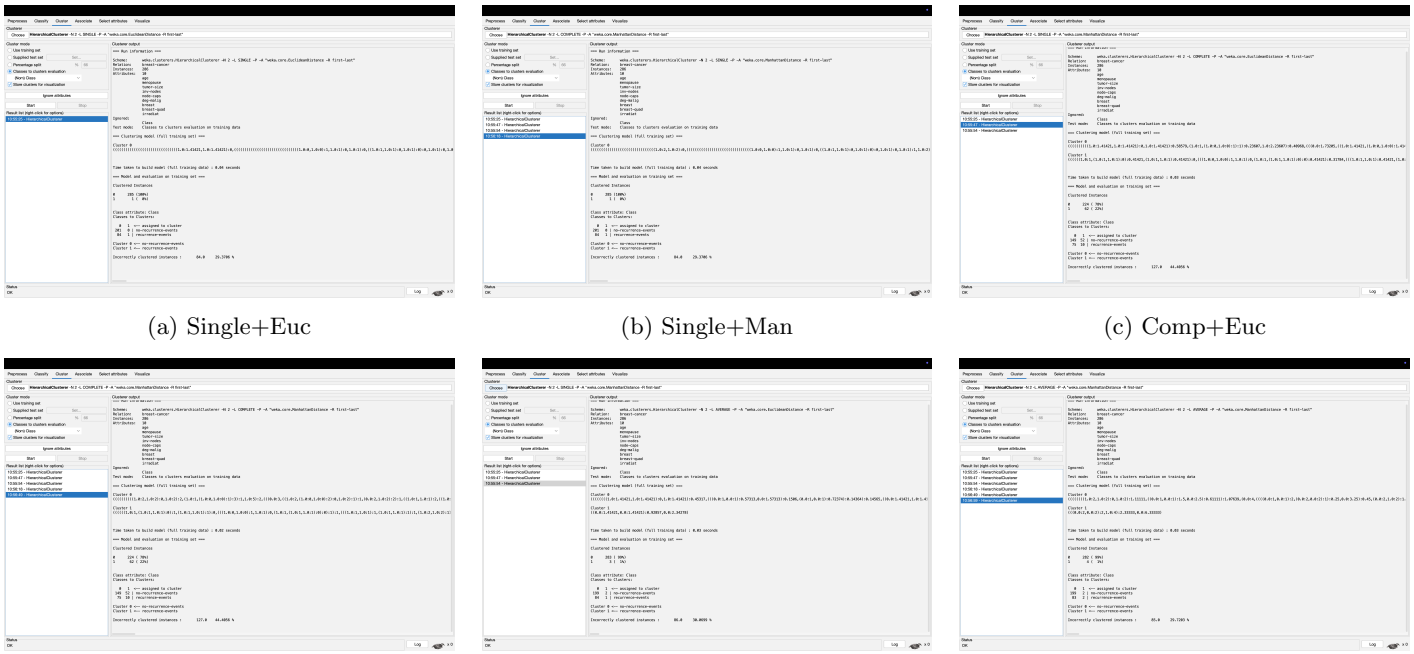
This section presents Hierarchical Agglomerative Clustering on `breast-cancer.arff` (2 clusters). Results are in Table 1.

Table 1: Summary of Accuracy for Hierarchical Clustering

Linkage	**Distance**	**Incorrect %**	**Accuracy**
Single	Euclidean	29.37%	**70.63%**
Single	Manhattan	29.37%	**70.63%**
Average	Manhattan	29.72%	70.28%
Average	Euclidean	30.07%	69.93%
Complete	Euclidean	44.41%	55.59%
Complete	Manhattan	44.41%	55.59%

Conclusion: Single Linkage performed best (70.63%), while Complete linkage performed worst.

Appendix: Weka Output



(a) Single+Euc (b) Single+Man (c) Comp+Euc

(d) Comp+Man (e) Avg+Euc (f) Avg+Man

Figure 1: Experimental results for all 6 combinations.

(b) FarthestFirst Clustering Stability

We evaluated **FarthestFirst** ($k = 2$) with seeds 1-10 (Table 2).

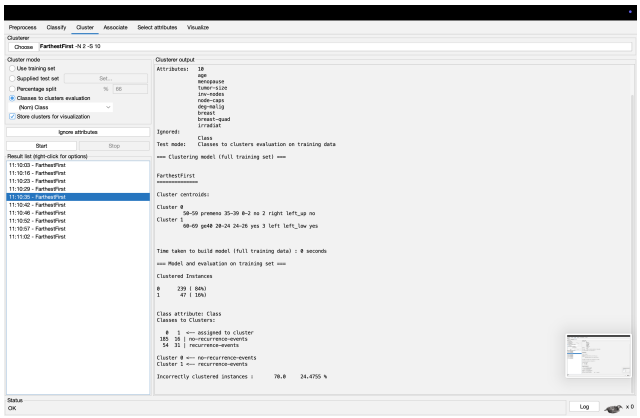
Conclusion

Range: Accuracy ranged from **51.05%** to **75.52%** (Variation: 24.5%).

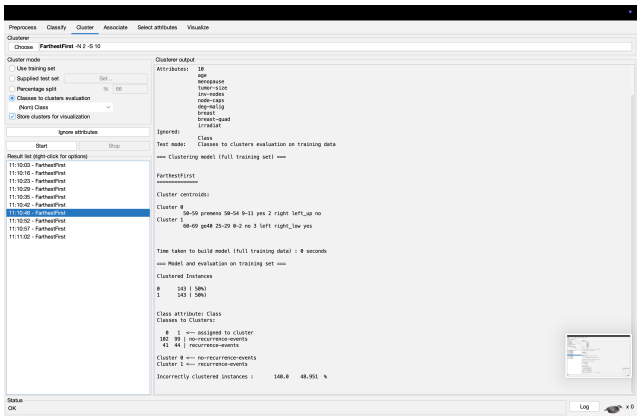
Stability: FarthestFirst is **highly sensitive to initialization**. Random seed selection leads to vastly different outcomes, making it unstable compared to Hierarchical Clustering.

Table 2: FarthestFirst Accuracy (Seeds 1-10)

Seed	Err %	Acc	Note	Seed	Err %	Acc	Note
1	34.27	65.73		6	29.37	70.63	
2	31.12	68.88		7	48.95	*51.05*	*Worst*
3	37.41	62.59		8	37.06	62.94	
4	30.42	69.58		9	31.47	68.53	
5	24.48	**75.52**	*Best*	10	24.83	75.17	



(a) Seed 5 (Best: 75.52%)



(b) Seed 7 (Worst: 51.05%)

Figure 2: Comparison showing high sensitivity to initialization.

(c) Compare the results

In this last section, we compare the unsupervised clustering algorithms (Q2a and Q2b) to a supervised classifier. We employed the decision tree algorithm **J48** (an implementation of C4.5) to the data in 10-fold cross-validation.

Results Comparison

The accuracy obtained from the J48 classifier is **75.52%**. Table 3 summarizes the best results from each method.

Table 3: Comparison of Clustering vs. Classification Accuracy

Method Type	Algorithm	Best Accuracy
Clustering (Unsupervised)	Hierarchical (Single Linkage)	70.63%
Clustering (Unsupervised)	FarthestFirst (Best Seed)	75.52%*
Classification (Supervised)	**J48 Decision Tree**	**75.52%**

*Note: FarthestFirst was highly unstable, with accuracy dropping as low as 51%.

Conclusion

Comparison:

The J48 classifier yields 75.52 % accuracy, which significantly improves over the stable clusters (Hierarchical at 70.63%). The "lucky" run of FarthestFirst (Seed 5) achieved a J48 accuracy but has to be discarded as an unreliable predicting model because of other very unstable runs.

Clustering's Capacity in Predicting Class Values:

We conjecture that clustering is **inadequate** when it comes to predicting the actual class value ("recurrence" and "no-recurrence") in this dataset.

That's because clustering is **unsupervised learning**; it just tries to group data by attribute similarity without keeping track of target labels. In the breast cancer data, the "actual clusters" of that data do not overlap exactly with what medical diagnoses a given sample receives.

In contrast, the J48 classifier applies **supervised learning,** which optimises for minimising the error of the model with respect to existing class labels, making it a better tool for this prediction.

Figure 3: Weka output for J48 Classifier showing 75.52% accuracy.

Question 3

(a) Apply DBSCAN

Applied the DBSCAN clustering algorithm to the `bmw-browsers.arff` dataset using the specified parameters: $\epsilon = 1.1$ and $MinPts = 10$. The algorithm identified two clusters (Cluster 0 and Cluster 1) along with some noise points.

Separating Attributes

To identify which attributes distinguish the two clusters, visualising the cluster assignments using Weka's visualization tool. Applied "Jitter" to separate overlapping points.

Founding that the attribute **3Series** provides a clear separation between the two clusters:

Cluster 1 (Red): Corresponds to visitors who viewed the **3Series** (Attribute value = 1).

Cluster 0 (Blue): Corresponds to visitors who did **not** view the 3Series (Attribute value = 0).

A similar inverse relationship was observed for the **M5** attribute, where Cluster 0 viewed the M5 and Cluster 1 did not. This indicates that the main dividing factor in the customer behavior is the choice between the M5 and the 3Series models.

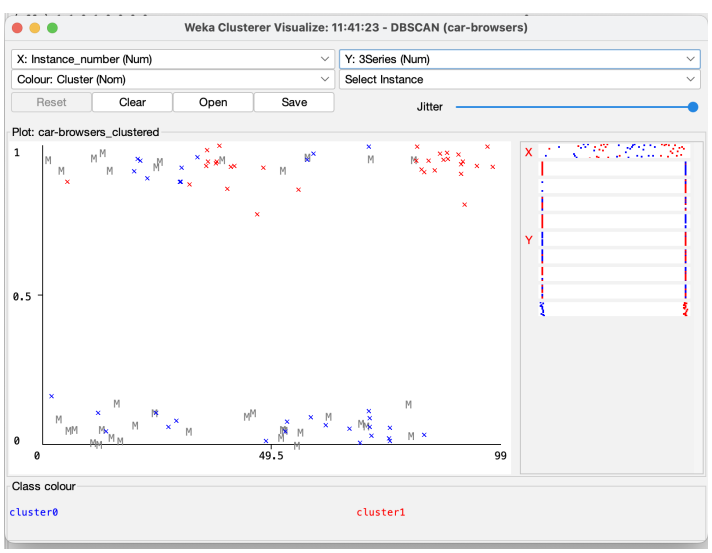


Figure 4: Visualization of DBSCAN results (Y-axis = 3Series). There is a clear separation: Cluster 1 (Red) is grouped at Y=1, while Cluster 0 (Blue) is grouped at Y=0.

(b) Apply the EM

Applied the EM (Expectation Maximization) algorithm to cluster the data into $k = 5$ clusters. Analyzed the cluster means to identify defining attributes (mean ≥ 0.9 or ≤ 0.1).

Analysis of Cluster Means (Seed = 1)

The output for Seed = 1 identified distinct customer behaviors. The significant attribute values are summarized below:

Cluster 0: Visitors who looked at the **M5** (1.0) but had a **Purchase** value of ≈ 0 (0.0058). Represents "M5 window shoppers".

Cluster 1: Visitors who went to the **Showroom** (1.0) and looked at both the **3Series** (1.0) and **Z4** (1.0), but ignored the **M5** (0.0).

Cluster 2: Visitors focused on the **3Series** (≈ 1.0) with no interest in the **M5** (0.0).

Cluster 3: Visitors who entered the **Dealership** (1.0) but had very low interest in the **3Series** (0.08).

Cluster 4: The most valuable group. They looked at the **M5** (1.0), used **Financing** (1.0), and had a **Purchase** probability of ≈ 1.0 (0.9965).

Consistency Across Seeds

We repeated the experiment with 2 and 3 as seeders.

Observation: The optimal clustering was **politely speaking, inconsistent**.
For instance, in seed 1, discovering the presence of a numerical cluster (Cluster 4) with an average purchase value$\approx$ 1.0 that we labelled "M5Buyers". But in later runs (e.g., Seed 2), there was no cluster whose Purchase mean was greater than 0.5, and the natural grouping "M5 + Financing" had to be shattered.

Conclusions: The EM algorithm may produce quite different models based on random seed on this dataset, presumably due to its sensitivity to initialisation.

(c) Find a k-means

Finally, applying the **SimpleKMeans** algorithm with $k = 5$ to the dataset. The resulting cluster centroids are shown in Figure 5(b).

Analysis of Cluster Centroids

We identified the significant dimensions for each cluster (centroid value ≥ 0.9 or ≤ 0.1):

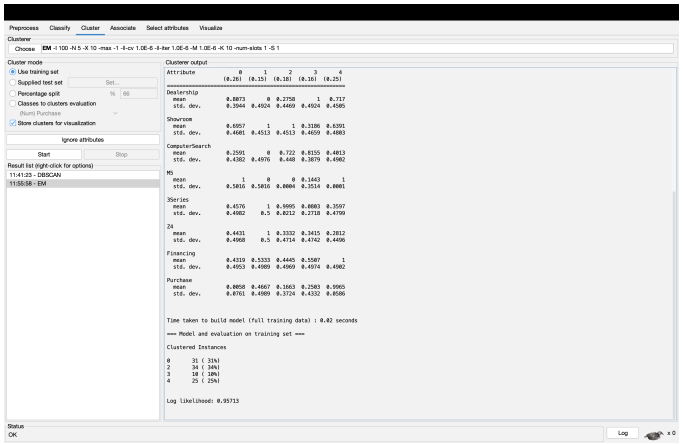
Cluster 0: Visitors who visited the **Dealership** (0.96) but definitely did not make a **Purchase** (0).

Cluster 1: Visitors highly interested in the **M5** (0.96) who did not perform a **ComputerSearch** (0).

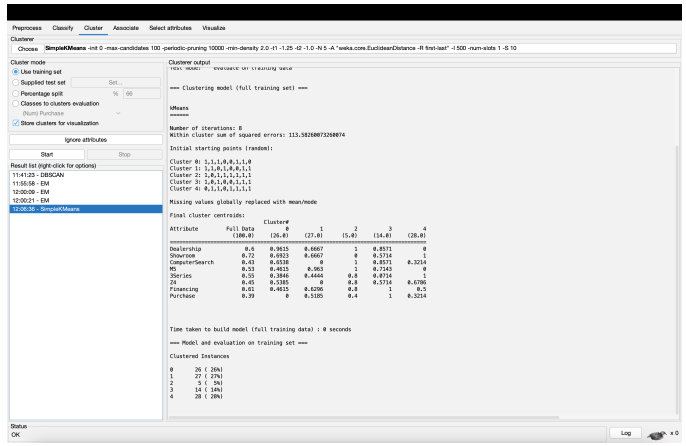
Cluster 2: Visitors who looked at the **M5** (1.0) and used **ComputerSearch** (1.0) and **Dealership** (1.0), but did not visit the **Showroom** (0).

Cluster 3 (Buyers): This group represents the successful conversions. They used **Financing** (1.0) and completed a **Purchase** (1.0). They had very low interest in the **3Series** (0.07).

Cluster 4 (3Series Fans): Visitors who visited the **Showroom** (1.0) and looked at the **3Series** (1.0), but ignored the **M5** (0) and **Dealership** (0).



(a) EM Clustering (Seed = 1). Note Cluster 4 (far right) shows high values for M5, Financing, and Purchase.



(b) SimpleKMeans ($k = 5$). Note the distinct profiles for M5 buyers vs. 3Series visitors.

Figure 5: Visual comparison of Weka outputs for EM (left) and SimpleKMeans (right) algorithms. Both methods consistently identify similar groups, such as the "M5 Buyer" group (EM Cluster 4 and K-Means Cluster 3).

Comparison with EM Results

The clusters found by SimpleKMeans are **highly consistent** with those found by the EM algorithm (in its stable Seed=1 state).

For instance:

> Both algorithms identified a distinct group of **"3Series Showroom Visitors"** (EM Cluster 1 vs. K-Means Cluster 4) who look at the 3Series but not the M5.
>
> Both algorithms identified a clear group of **"Financing Buyers"** (EM Cluster 4 vs. K-Means Cluster 3) characterized by high values for Financing and Purchase.

This consistency suggests that these customer profiles (e.g., "M5 Buyer" vs. "3Series Window Shopper") are strong, natural patterns in the data that are robust across different clustering techniques.